Phillip Macias

Data Scientist — Statistical Modeling & Experimentation, Mountain View, CA

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Summary

Quantitative scientist with 10+ years turning large-scale simulations and data into decisions, including a survey-design recommendation adopted by NASA for a $4.3B flagship mission. Recent work applies experimentation, causal inference, Bayesian inference, and machine learning to real A/B-test, gameplay, and environmental data.

Skills

- **Computation:** Python (NumPy, SciPy, pandas, PyMC, ArviZ, scikit-learn, Matplotlib, Jupyter), SQL, DuckDB, Git, HPC (Slurm, large parameter sweeps), agentic AI workflows (Claude Code, LLM APIs)
- **Statistics/Machine Learning:** A/B testing (hypothesis tests, confidence intervals, power/MDE, multiple-testing correction), causal inference (difference-in-differences, event studies, placebo tests), Bayesian & hierarchical modeling (MCMC/NUTS), regression & classification (XGBoost, Explainable Boosting Machines), model explainability (SHAP), feature engineering

Experience and Projects

Spirometry — Causal Effects of Game Balance Patches (Self-Directed Project) **2026**

- Estimated how 43 numeric card changes in *Slay the Spire 2*'s early-access patches moved player pick rates, using a difference-in-differences design (changed card vs. the same character's unchanged cards) on ∼1.1M community-uploaded runs, with offer-balance, pre-trend, and robustness checks.
- Ran an 860-fit placebo test that showed the clustered standard errors were too small (placebo z spread 4.2 vs. 1 expected), then rescaled them: players responded clearly to the first patch (13 of 26 changes) but barely to the second (2 of 17).

Clickworthy — Meta-Analysis of 3,800+ Real A/B Tests (Self-Directed Project) **2026**

- Audited 3,823 randomized headline experiments (Upworthy Research Archive) with classical inference: the median test could only detect an ∼83% relative lift, and "significant" tests fell from 44% to 34% under Benjamini–Hochberg FDR control.
- Measured winner's-curse inflation in reported winners via split-sample re-estimation, and showed by A/A simulation that peeking at live results inflates the false-positive rate from 5% to 25%.
- Built a hierarchical Bayesian model in PyMC (non-centered, NUTS, zero divergences) that partially pools arm effects across all tests, shrinking noisy estimates in proportion to their sample size.

Flashpoint — Wildfire Escalation Classifier (Self-Directed Project) **2026**

- Built an end-to-end pipeline predicting whether an active wildfire escalates from its first 1–2 days of growth plus weather, topography, and vegetation features; compared XGBoost, EBM, and a CNN, with EBM giving the highest escalation-recall (0.95) across held-out years.

Roman Space Telescope Planning (Postdoctoral Researcher, UC Santa Cruz) **2020 – 2026**

- Independently ran ∼500 end-to-end simulation campaigns evaluating candidate survey designs (depth, cadence, filters) against a NASA requirement for a 10x gain in dark-energy constraining power, reducing each design to a single comparable figure-of-merit.
- Synthesized simulation results into a recommended survey design adopted by NASA, which exceeded the requirement (FoM 340 vs. 326 target) and directly shaped the observing strategy for the flagship dark-energy survey of *Roman*, a $4.3B NASA mission that launched in August 2026.

Graduate Research (NSF Graduate Research Fellow, UC Santa Cruz) **2012 – 2019**

- Ran large hydrodynamic simulations of accretion flows (5 co-authored papers) and used high-resolution stellar spectra to rule out 2 leading heavy-element production channels (2 first-author papers).

21 peer-reviewed publications, 700 citations, 20+ students mentored across astronomy research.

Education

Ph.D., Astronomy & Astrophysics — University of California, Santa Cruz *2019*

M.S., Astronomy & Astrophysics — University of California, Santa Cruz *2015*

B.S., Physics (Highest Honors) — University of California, Santa Barbara *2012*